

# Online Job Advertisements

Data dictionary



# Online Job Advertisements Structural Metadata

Web Intelligence Hub - Eliane Gotuzzo (eliane.gotuzzo@ext.ec.europa.eu)

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# 1 Overview

This document describes the structure of the dataflows WIH-OJA and WIH-OJA-NLP, in particular as they are made available via database access. It includes the record structure of the several tables available under these two dataflows and the data dictionary.

## 2 WIH-OJA methodology

For detailed information on the methodology, statistical presentation, data descriptions, classifications used, data coverage, statistical concepts and definitions, release policies, and other statistical elements, please refer to the WIH-OJA reference metadata file (ESMS): [https://ec.europa.eu/eurostat/cache/metadata/Annexes/isoc\\_sk\\_oja\\_esmsip2\\_an\\_1.pdf](https://ec.europa.eu/eurostat/cache/metadata/Annexes/isoc_sk_oja_esmsip2_an_1.pdf).

In addition, members who already use OJA data and have access to it can find further information on the Web Intelligence Hub wiki space: <https://webgate.ec.europa.eu/fpfis/wikis/x/GIuflg>

## 3 Access to the WIH-OJA microdata

### 3.1 Data availability

OJA data for scientific purposes are available in the following form:

- “Scientific-use datasets” – available through a database connection accessible via a remote data lab, a web-based environment providing data science tools, including JupyterLab and RStudio, where users can query the data using SQL.

Scientific-use datasets are available for the following dataflows:

- WIH-OJA: Online Job Advertisements classified data
- WIH-OJA-NLP: Job titles and job descriptions from OJAs

### 3.2 Use OJA microdata via the WIH DataLab

Users who already have access to OJA data can find further information on how to query the OJA microdata in the documentation on the Web Intelligence Hub wiki space at: <https://webgate.ec.europa.eu/fpfis/wikis/x/3QHkbw>

## 4 Dataflow: WIH-OJA

### 4.1 OJA Blended Tables

#### 4.1.1 Description

Tables with all OJA variables, including variables with several values for the same OJA (skills, location, source). As a result, an OJA can be present in several records, one for each combination of each skill and each location identified in the OJA. It includes OJAs for EU Member-States and the UK.

#### 4.1.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

#### 4.1.3 Tables Released

- wih\_oja\_blended\_v2\_2021q4\_r20220224
- wih\_oja\_blended\_v2\_2022q4\_r20230213
- wih\_oja\_blended\_v2\_2023q4\_r20240226
- wih\_oja\_blended\_v2\_2024q3\_r20241113
- wih\_oja\_blended\_v2\_2024q4\_r20250313
- wih\_oja\_blended\_v2\_2025q3\_r20260216
- wih\_oja\_blended\_v2\_2025q4\_r20260313
- wih\_oja\_blended\_v2\_2026q1\_r20260429
- wih\_oja\_blended\_v2\_2026q2\_r20260722
- wih\_oja\_blended

#### 4.1.4 Record Structure

- oja\_id
- first\_active\_date
- first\_active\_year
- first\_active\_month
- first\_active\_day
- last\_active\_date
- last\_active\_year
- last\_active\_month
- last\_active\_day
- language
- occupation4d\_id
- occupation4d
- occupation3d\_id
- occupation3d
- occupation2d\_id
- occupation2d
- occupation1d\_id
- occupation1d
- skill\_id
- skill
- skill\_hier3\_id
- skill\_hier3
- skill\_hier2\_id
- skill\_hier2
- skill\_hier1\_id
- skill\_hier1
- skill\_hier0\_id
- skill\_hier0
- city\_id
- city
- nuts3\_id
- nuts3
- nuts2\_id
- nuts2
- nuts1\_id
- nuts1
- country\_id
- country

- contract\_id
- contract
- education\_id
- education
- economic\_activity2d\_id
- economic\_activity2d
- economic\_activity1d\_id
- economic\_activity1d
- activity\_sector\_id
- activity\_sector
- salary\_id
- salary
- working\_time\_id
- working\_time
- experience\_id
- experience
- source\_id
- source\_category
- source\_category\_id
- source\_country
- source\_stability

## 4.2 OJA Main Tables

### 4.2.1 Description

Tables including all OJA variables for which there is only one value possible for each OJA. It excludes skills and location.

### 4.2.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.2.3 Tables Released

- wih\_oja\_main\_v2\_2021q4\_r20220224
- wih\_oja\_main\_v2\_2022q4\_r20230213
- wih\_oja\_main\_v2\_2023q4\_r20240226
- wih\_oja\_main\_v2\_2024q3\_r20241113
- wih\_oja\_main\_v2\_2024q4\_r20250313
- wih\_oja\_main\_v2\_2025q3\_r20260216
- wih\_oja\_main\_v2\_2025q4\_r20260313
- wih\_oja\_main\_v2\_2026q1\_r20260429
- wih\_oja\_main\_v2\_2026q2\_r20260722
- wih\_oja\_main

### 4.2.4 Record Structure

- oja\_id
- first\_active\_date

- last\_active\_date
- language
- contract\_id
- education\_id
- salary\_id
- working\_time\_id
- experience\_id
- occupation4d\_id
- occupation3d\_id
- occupation2d\_id
- occupation1d\_id
- economic\_activity2d\_id
- economic\_activity1d\_id

## 4.3 OJA Skills Tables

### 4.3.1 Description

Tables with information on the skills required in each OJA. Each skill required is an individual record. So OJAs normally are present in several records.

### 4.3.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.3.3 Tables Released

- wih\_oja\_skill\_v1\_2021q4\_r20220224
- wih\_oja\_skill\_v1\_2022q4\_r20230213
- wih\_oja\_skill\_v1\_2023q4\_r20240226
- wih\_oja\_skill\_v1\_2024q3\_r20241113
- wih\_oja\_skill\_v1\_2024q4\_r20250313
- wih\_oja\_skill\_v1\_2025q3\_r20260216
- wih\_oja\_skill\_v1\_2025q4\_r20260313
- wih\_oja\_skill\_v1\_2026q1\_r20260429
- wih\_oja\_skill\_v1\_2026q2\_r20260722
- wih\_oja\_skill

### 4.3.4 Record Structure

- oja\_id
- skill\_id
- skill\_hier3\_id
- skill\_hier2\_id
- skill\_hier1\_id
- skill\_hier0\_id

## 4.4 OJA codelist tables

### 4.4.1 Description

OJA codelist tables

### 4.4.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.4.3 Tables Released

- codelist\_wih\_oja\_v1
- codelist\_wih\_oja

### 4.4.4 Record Structure

- code
- label
- variable

## 4.5 OJA location codelist tables

### 4.5.1 Description

Location codelist tables

### 4.5.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.5.3 Tables Released

- codelist\_wih\_oja\_city\_v1\_2021q4\_r20220224
- codelist\_wih\_oja\_city\_v1\_2022q4\_r20230213
- codelist\_wih\_oja\_city\_v1\_2023q4\_r20240226
- codelist\_wih\_oja\_city\_v2\_1\_2024q3\_r20241113
- codelist\_wih\_oja\_city\_v2\_1\_2024q4\_r20250313
- codelist\_wih\_oja\_city\_v2\_2025q3\_r20260216
- codelist\_wih\_oja\_city\_v2\_2025q4\_r20260313
- codelist\_wih\_oja\_city\_v2\_2026q1\_r20260429
- codelist\_wih\_oja\_city\_v2\_2026q2\_r20260722
- codelist\_wih\_oja\_city

### 4.5.4 Record Structure

- Please check the table specific record structure

## 4.6 OJA skill codelist tables

### 4.6.1 Description

OJA ESCO skill codelist tables

### 4.6.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.6.3 Tables Released

- codelist\_wih\_oja\_skill\_v1\_2021q4\_r20220224
- codelist\_wih\_oja\_skill\_v2\_2022q4\_r20230213
- codelist\_wih\_oja\_skill\_v3\_2023q4\_r20240226
- codelist\_wih\_oja\_skill\_v3\_2024q3\_r20241113
- codelist\_wih\_oja\_skill\_v3\_2024q4\_r20250313
- codelist\_wih\_oja\_skill\_v3\_2025q3\_r20260216
- codelist\_wih\_oja\_skill\_v3\_2025q4\_r20260313
- codelist\_wih\_oja\_skill\_v3\_2026q1\_r20260429
- codelist\_wih\_oja\_skill\_v3\_2026q2\_r20260722
- codelist\_wih\_oja\_skill

### 4.6.4 Record Structure

- Please check the table specific record structure

## 4.7 OJA Source Link Tables

### 4.7.1 Description

Tables with information on the sources where OJAs were found. Each source where an OJA was found is an individual record. So OJAs may be present in several records, one for each source where it was found.

### 4.7.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.7.3 Tables Released

- wih\_oja\_source\_link\_v1\_2021q4\_r20220224
- wih\_oja\_source\_link\_v1\_2022q4\_r20230213
- wih\_oja\_source\_link\_v1\_2023q4\_r20240226
- wih\_oja\_source\_link\_v1\_2024q3\_r20241113
- wih\_oja\_source\_link\_v1\_2024q4\_r20250313
- wih\_oja\_source\_link\_v1\_2025q3\_r20260216
- wih\_oja\_source\_link\_v1\_2025q4\_r20260313
- wih\_oja\_source\_link\_v1\_2026q1\_r20260429

- wih\_oja\_source\_link\_v1\_2026q2\_r20260722
- wih\_oja\_source\_link

#### 4.7.4 Record Structure

- oja\_id
- source\_id
- source\_category\_id
- source\_country
- source\_stability

### 4.8 OJA Location Tables

#### 4.8.1 Description

Location tables

#### 4.8.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

#### 4.8.3 Tables Released

- wih\_oja\_location\_v1\_2021q4\_r20220224
- wih\_oja\_location\_v1\_2022q4\_r20230213
- wih\_oja\_location\_v1\_2023q4\_r20240226
- wih\_oja\_location\_v1\_2024q3\_r20241113
- wih\_oja\_location\_v1\_2024q4\_r20250313
- wih\_oja\_location\_v1\_2025q3\_r20260216
- wih\_oja\_location\_v1\_2025q4\_r20260313
- wih\_oja\_location\_v1\_2026q1\_r20260429
- wih\_oja\_location\_v1\_2026q2\_r20260722
- wih\_oja\_location

#### 4.8.4 Record Structure

- oja\_id
- city\_id
- nuts3\_id
- nuts2\_id
- nuts1\_id
- country\_id

### 4.9 OJA Date Tables

#### 4.9.1 Description

Helper table to convert the dates present in WIH-OJA data from unix epoch format to year, month, day.

#### 4.9.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

#### 4.9.3 Tables Released

- wih\_oja\_date\_v1
- wih\_oja\_date

#### 4.9.4 Record Structure

- date
- year
- month
- day

### 4.10 OJA EFTA Blended Tables

#### 4.10.1 Description

Tables with all OJA variables, including variables with several values for the same OJA (skills, location). As a result, an OJA can be present in several records, one for each combination of each skill and each location identified in the OJA. It includes OJAs for EFTA Member-States (NO, IS, CH, LI).

#### 4.10.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

#### 4.10.3 Tables Released

- wih\_oja\_blended\_efta\_v0\_1\_2023q4\_r20240226
- wih\_oja\_blended\_efta\_v0\_1\_2024q2\_r20240814
- wih\_oja\_blended\_efta\_v0\_1\_2024q3\_r20241113
- wih\_oja\_blended\_efta\_v0\_1\_2024q4\_r20250313
- wih\_oja\_blended\_efta

#### 4.10.4 Record Structure

- oja\_id
- first\_active\_date
- first\_active\_year
- first\_active\_month
- first\_active\_day
- last\_active\_date
- last\_active\_year
- last\_active\_month
- last\_active\_day
- language

- occupation4d\_id
- occupation4d
- occupation3d\_id
- occupation3d
- occupation2d\_id
- occupation2d
- occupation1d\_id
- occupation1d
- skill\_id
- skill
- skill\_hier3\_id
- skill\_hier3
- skill\_hier2\_id
- skill\_hier2
- skill\_hier1\_id
- skill\_hier1
- skill\_hier0\_id
- skill\_hier0
- city\_id
- city
- nuts3\_id
- nuts3
- nuts2\_id
- nuts2
- nuts1\_id
- nuts1
- country\_id
- country
- contract\_id
- contract
- education\_id
- education
- economic\_activity2d\_id
- economic\_activity2d
- economic\_activity1d\_id
- economic\_activity1d
- activity\_sector\_id
- activity\_sector
- salary\_id
- salary
- working\_time\_id
- working\_time
- experience\_id
- experience
- source\_id
- source\_category
- source\_category\_id
- source\_country
- source\_stability

## 4.11 OJA Job Descriptions

### 4.11.1 Description

OJA NLP tables

### 4.11.2 Databases

- wih\_oja\_nlp

### 4.11.3 Tables Released

- wih\_oja\_nlp\_v2\_2022\_r20230315
- wih\_oja\_nlp\_v2\_2023\_r20240226
- wih\_oja\_nlp\_v2\_2024\_r20250625
- wih\_oja\_nlp\_v2\_2025\_r20260506

### 4.11.4 Record Structure

- oja\_id
- source
- site
- description
- description\_processed
- title
- title\_processed
- country\_id
- country
- first\_active\_year
- first\_active\_month
- lang
- occupation4d\_id
- occupation4d
- occupation3d\_id
- occupation3d
- occupation2d\_id
- occupation2d
- occupation1d\_id
- occupation1d
- economic\_activity2d\_id
- economic\_activity2d
- economic\_activity1d\_id
- economic\_activity1d
- contract\_id
- contract
- education\_id
- education
- salary\_id
- salary
- working\_time\_id
- working\_time
- experience\_id
- experience

## 4.12 OJA NLP tables

### 4.12.1 Description

OJA NLP tables 2021

### 4.12.2 Databases

- wih\_oja\_nlp

### 4.12.3 Tables Released

- wih\_oja\_nlp\_v1\_2021\_r20221217

### 4.12.4 Record Structure

- oja\_id
- description
- title
- country\_id
- country
- first\_active\_year
- first\_active\_month
- lang
- occupation4d\_id
- occupation4d
- occupation3d\_id
- occupation3d
- occupation2d\_id
- occupation2d
- occupation1d\_id
- occupation1d
- economic\_activity2d\_id
- economic\_activity2d
- economic\_activity1d\_id
- economic\_activity1d
- contract\_id
- contract
- education\_id
- education
- salary\_id
- salary
- working\_time\_id
- working\_time
- experience\_id
- experience

## 4.13 OJA NLP occupation tables

### 4.13.1 Description

NLP-based occupation classification tables in the `wih_oja_nlp` database. Includes ISCO-coded occupations inferred from job titles and descriptions using various classification versions and token-matching methods. Version history includes multiple releases (v1–v3), covering updates and refinements to the classification approach.

### 4.13.2 Databases

- `wih_oja_nlp`

### 4.13.3 Tables Released

- `wih_oja_nlp_occ_v3_2022_r20231201`
- `wih_oja_nlp_occ_v3_2023_r20240226`
- `wih_oja_nlp_occ_v3_2024_r20250625`
- `wih_oja_nlp_occ_v3_2025_r20260506`

### 4.13.4 Record Structure

- `oja_id`
- `description`
- `title`
- `first_active_year`
- `lang`
- `country_id`
- `country`
- `nuts2_id`
- `nuts2`
- `occupation4d_id`
- `occupation4d`
- `occupation3d_id`
- `occupation3d`
- `occupation2d_id`
- `occupation2d`
- `occupation1d_id`
- `occupation1d`
- `economic_activity2d_id`
- `economic_activity2d`
- `economic_activity1d_id`
- `economic_activity1d`
- `source`
- `site`

## 4.14 OJA NLP skill tables

### 4.14.1 Description

NLP skills tables with ESCO level 3 skills and hierarchy codes (levels 0–3).

#### 4.14.2 Databases

- wih\_oja\_nlp

#### 4.14.3 Tables Released

- wih\_oja\_nlp\_skills\_v3\_2022\_r20231201
- wih\_oja\_nlp\_skills\_v3\_2023\_r20240226
- wih\_oja\_nlp\_skills\_v3\_2024\_r20250625
- wih\_oja\_nlp\_skills\_v3\_2025\_r20260506

#### 4.14.4 Record Structure

- oja\_id
- description
- title
- first\_active\_year
- first\_active\_month
- lang
- country\_id
- country
- skill\_id
- skill
- skill\_hier3\_id
- skill\_hier3
- skill\_hier2\_id
- skill\_hier2
- skill\_hier1\_id
- skill\_hier1
- skill\_hier0\_id
- skill\_hier0

### 4.15 OJA NLP translated tables

#### 4.15.1 Description

NLP translations of titles and descriptions, including EN versions and classified variables.

#### 4.15.2 Databases

- wih\_oja\_nlp

#### 4.15.3 Tables Released

- wih\_oja\_nlp\_v2\_1\_2022\_r20240722

#### 4.15.4 Record Structure

- oja\_id
- lang
- title\_en
- description\_en
- source
- site
- description\_processed
- title\_processed
- country\_id
- country
- description
- title
- first\_active\_year
- first\_active\_month
- occupation4d\_id
- occupation4d
- occupation3d\_id
- occupation3d
- occupation2d\_id
- occupation2d
- occupation1d\_id
- occupation1d
- economic\_activity2d\_id
- economic\_activity2d
- economic\_activity1d\_id
- economic\_activity1d
- contract\_id
- contract
- education\_id
- education
- salary\_id
- salary
- working\_time\_id
- working\_time
- experience\_id
- experience

### 4.16 Codelist Tables

#### 4.16.1 Description

OJA codelist tables

#### 4.16.2 Databases

- wih\_oja\_versioned
- wih\_oja\_latest

### 4.16.3 Tables Released

- codelist\_wih\_oja\_v1
- codelist\_wih\_oja

### 4.16.4 Record Structure

- code
- label
- variable

## 5 OJA data dictionary

This OJA data dictionary user guide focuses on the concepts, variables definition and data structure.

Table 1: OJA Data dictionary

| variable_name      | description                                                                                                                                                                                                                                                                                                                                                                                  | classification | format                                                                                       |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------|
| oja_id             | Identifier of the online job advertisement. Please note that this variable identifies the job advertisement and not job vacancies. The same oja_id may refer to different locations, sources and skills.                                                                                                                                                                                     |                | 9 digital characters                                                                         |
| first_active_date  | Provides the date when the posting is first detected for a specific source. In cases, where the publishing date is extracted from the advertisement, this date is provided as grab date, otherwise the date when the detection is done is provided.                                                                                                                                          |                | The date is presented in unix epoch format (5 digits), measured in days from January 1, 1970 |
| first_active_year  | Year extracted from the variable first_active_date. Derived variable from first_active_date. See <a href="https://en.wikipedia.org/wiki/Date_format_by_country">https://en.wikipedia.org/wiki/Date_format_by_country</a> for explanation of the dates coding.                                                                                                                                |                | yyyy                                                                                         |
| first_active_month | Month extracted from the variable first_active_date. Derived variable from first_active_date..                                                                                                                                                                                                                                                                                               |                | m                                                                                            |
| first_active_day   | Day of the month extracted from variable first_active_date. Derived variable from first_active_date.                                                                                                                                                                                                                                                                                         |                | dd                                                                                           |
| last_active_date   | Provides the date when the posting disappears from the source or the actual expiration of the posting, if stated in the advertisement, measured in days from January 1, 1970. As often information on expire date is not available, it is imputed to a standard value (evaluated through interquartile range calculation by country, occupation and industry), if exceeding admitted values. |                | the date is in unix epoch format.                                                            |
| last_active_year   | Year extracted from variable last_active_date. Derived variable from last_active_date                                                                                                                                                                                                                                                                                                        |                | yyyy                                                                                         |
| last_active_month  | Month extracted from variable last_active_date. Derived variable from last_active_date.                                                                                                                                                                                                                                                                                                      |                | m                                                                                            |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| last_active_day | Day extracted from variable last_active_date. Derived variable from last_active_date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                | d                  |
| language        | ISO 639-1 language code in which the advertisement is written. Languages belong to a list of 29 European languages.<br>ISO 639-1:2002, Codes for the representation of names of languages—Part 1: Alpha-2 code, is the first part of the ISO 639 series of international standards for language codes. Part 1 covers the registration of two-letter codes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | ISO 639-1:2002 | Alpha-2 code       |
| occupation4d_id | ISCO level 4 code of occupation required in the job<br>divertissement in International standard classification of occupations 2008, 4 digits – ISCO-08<br>( <a href="https://isco.ilo.org/en/">https://isco.ilo.org/en/</a> ).<br>Occupations in ESCO are structured through their mapping to the International Standard Classification of Occupations (ISCO-08), which has been developed by the International Labour Organisation (ILO). The ESCO occupations and their ISCO-08 hierarchy make up the ESCO occupations pillar.<br>ISCO-08 provides the top four levels while ESCO occupations provide the fifth and lower levels. Each ESCO occupation is assigned to one ISCO-08 unit group (even if they are not directly related to it, e.g. if they are at level six or seven). The ESCO - classification levels and taxonomy can be found here:<br><a href="https://ec.europa.eu/esco/portal/occupation">https://ec.europa.eu/esco/portal/occupation</a><br>Definition:<br>Occupation is a "set of jobs whose main tasks and duties are characterised by a high degree of similarity"<br>(International Labour Organisation (ILO), International Standard Classification of Occupations (ISCO-08) – Conceptual Framework,<br><a href="https://ec.europa.eu/esco/portal/document/en/0a89839c-098d-4e34-846c-54cbd5684d24">https://ec.europa.eu/esco/portal/document/en/0a89839c-098d-4e34-846c-54cbd5684d24</a> ). | ISCO-08        | 4 digits – ISCO-08 |
| occupation4d    | International standard classification of occupations 2008, 4 digits – ISCO-08 ( <a href="https://isco.ilo.org/en/">https://isco.ilo.org/en/</a> )<br>Name of the occupation in the ISCO-08 classification.<br>Occupations can be used as job titles.<br>Usually, the employers title the vacancy notice with the name of an occupation<br>( <a href="https://ec.europa.eu/esco/portal/escopedia/Occupation">https://ec.europa.eu/esco/portal/escopedia/Occupation</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | ISCO-08        | string             |
| occupation3d_id | ISCO level 3 code of occupation required in the job<br>divertissement in International standard classification of occupations 2008, 3 digits – ISCO-08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ISCO-08        | 3 digits – ISCO-08 |
| occupation3d    | International standard classification of occupations 2008, 3 digits – ISCO-08<br>Name of the occupation in the ISCO-08 classification.<br>Occupations can be used as job titles. Usually, the employers title the vacancy notice with the name of an occupation<br>( <a href="https://ec.europa.eu/esco/portal/escopedia/Occupation">https://ec.europa.eu/esco/portal/escopedia/Occupation</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | ISCO-08        | string             |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |                                      |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------------------------------------|
| occupation2d_id | ISCO level 2 code of occupation required in the job<br>divertissement in International standard classification of<br>occupations 2008, 2 digits – ISCO-08                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ISCO-08 | 2 digits –<br>ISCO-08                |
| occupation2d    | International standard classification of occupations 2008, 2<br>digits – ISCO-08<br>Name of the occupation in the ISCO-08 classification.<br>Occupations can be used as job titles. Usually, the<br>employers title the vacancy notice with the name of an<br>occupation (<br><a href="https://ec.europa.eu/esco/portal/escopedia/Occupation">https://ec.europa.eu/esco/portal/escopedia/Occupation</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ISCO-08 | string                               |
| occupation1d_id | International standard classification of occupations 2008, 1<br>digit – ISCO-08<br>ISCO level 1 code of occupation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ISCO-08 | 1 digit –<br>ISCO-08                 |
| occupation1d    | International standard classification of occupations 2008, 1<br>digit – ISCO-08<br>Name of the occupation in the ISCO-08 classification.<br>Occupations can be used as job titles. Usually, the<br>employers title the vacancy notice with the name of an<br>occupation<br>( <a href="https://ec.europa.eu/esco/portal/escopedia/Occupation">https://ec.europa.eu/esco/portal/escopedia/Occupation</a> ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ISCO-08 | string                               |
| skill_id        | ESCO level 3 code of skill required in the job<br>divertissement.<br>The skills pillar of ESCO contains 13,485 concepts<br>structured in a hierarchy which contains four<br>sub-classifications, that can be found here:<br><a href="https://ec.europa.eu/esco/portal/skill">https://ec.europa.eu/esco/portal/skill</a><br>It can also be found in the blended and skill tables. Please<br>check OJA tables.<br>Commission published ESCOv1<br>( <a href="https://ec.europa.eu/esco/portal/escopedia/ESCO_v1">https://ec.europa.eu/esco/portal/escopedia/ESCO_v1</a> ) -<br>as Linked Open Data<br>( <a href="https://open-data.europa.eu/en/data">https://open-data.europa.eu/en/data</a> ). Thus, ESCO can<br>be connected to:<br>classification systems (ISCO-08, national classifications,<br>ISCED-F 2013, ...), controlled vocabularies and<br>frameworks (European Qualifications Framework (EQF),<br>Digital Competence Framework (DigComp), databases,<br>syntactical standards or tool, that make use of ESCO to<br>provide services.<br>Definition:<br>Skill is defined as "the ability to carry out the tasks and<br>duties of a given job".<br>For the purposes of ISCO-08, two dimensions of skill are<br>used to arrange occupations into groups: skill level and<br>skill specialization.<br>Skill level is defined as a function of the complexity and<br>range of tasks and duties to be performed in an occupation. |         | SCO v1<br> ESCOv1_(4 or<br>5 digits) |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |    |                           |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------|
| skill           | Name/label of the skill - ESCO level 3 - required in the job advertisement.<br>Available only in the table 'ft_skill_analysis_en', for versions up to V9.<br>It can also be found in the blended and skill tables. Please check OJA tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    | string                    |
| skill_uri       | Classification: ESCO v1 Concept URI.<br>Description: Official ESCO URI for the skill concept. Used for semantic linking across datasets.<br>Format: URL (e.g., <a href="http://data.europa.eu/esco/skill/...">http://data.europa.eu/esco/skill/...</a> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | NA | NA                        |
| skill_hier3_id  | The ESCO skill/competences (2nd pillar) hierarchy code level 3.<br>The ESCO skills pillar distinguishes between:<br>i) skill/competence concepts<br>ii) knowledge concepts → skill type.<br>There is no distinction between skills and competences.<br>Each of these concepts comes with one preferred term and a number of non-preferred terms in each of the 27 ESCO languages.<br>The ESCO skills concepts are structured in a 4 sub-classifications hierarchy targeting different types of knowledge and skill/competence (KSC) concepts:<br>K - Knowledge S - Skills A - Attitudes and values L - Language skills and knowledge<br>The ESCO skill hierarchy - <a href="https://ec.europa.eu/esco/portal/skill">https://ec.europa.eu/esco/portal/skill</a> - is partially based on elements of the existing classification of O*Net and the Canadian skill & knowledge glossary. For more information click here.<br>It can be found in the blended and skill tables. Please check OJA tables. |    | URL link<br>(Concept URI) |
| skill_hier3     | Name/label of the skill - according to the ESCO skill/competences hierarchy level 3 - specified in the job advertisement.<br>It can also be found in the blended and skill tables. Please check OJA tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    | string                    |
| skill_hier3_uri | Classification: ESCO v1 level 3 skill domain (partially aligned with ISCED-F fields).<br>Description: Broad grouping of skills.<br>Format: URL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NA | NA                        |
| skill_hier2_id  | The ESCO skill/competences (2nd pillar) hierarchy code level 2.<br>Available only in the table 'ft_skill_analysis_en', for versions up to V9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |    | URL link<br>(Concept URI) |
| skill_hier2     | Name/label of the skill - according to the ESCO skill/competences hierarchy level 2 - specified in the job advertisement.<br>It can also be found in the blended and skill tables. Please check OJA tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |    | string                    |
| skill_hier2_uri | Classification: ESCO v1 level 2 sub-domain.<br>Description: Intermediate grouping of skills within a domain.<br>Format: URL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | NA | NA                        |

|                  |                                                                                                                                                                                                                                                                                                                                |    |                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------|
| skill_hier1_id   | The ESCO skill/competences (2nd pillar) hierarchy code level 1.<br>Available only in the table 'ft_skill_analysis_en', for versions up to V9.                                                                                                                                                                                  |    | URL link<br>(Concept URI) |
| skill_hier1      | Name/label of the skill - according to the ESCO skill/competences hierarchy level 1 - specified in the job advertisement.<br>It can also be found in the blended and skill tables. Please check OJA tables.                                                                                                                    |    | string                    |
| skill_hier1_uri  | Classification: ESCO v1 level 1 skill domain (partially aligned with ISCED-F fields).<br>Description: Broad grouping of skills.<br>Format: URL.                                                                                                                                                                                | NA | NA                        |
| skill_hier0_id   | The ESCO skill/competences (2nd pillar) sub-classification code (url link).<br>It can also be found in the blended and skill tables. Please check OJA tables.                                                                                                                                                                  |    | URL link<br>(Concept URI) |
| skill_hier0      | Name/label of the sub-classification, according to the ESCO skill/competences hierarchy (Knowledge, Skills, Attitudes and values, Language skills and knowledge) that specified in the job advertisement.<br>It can also be found in the blended and skill tables. Please check OJA tables.                                    |    | string                    |
| skill_hier0_uri  | The root level of the ESCO Skills Hierarchy (URI)                                                                                                                                                                                                                                                                              | NA | NA                        |
| skill_obsolete   | Classification: Boolean flag (Y/N).<br>Description: Indicates if the skill has been deprecated in ESCO v1.0.1.                                                                                                                                                                                                                 | NA | NA                        |
| skill_digcomp    | Classification: Boolean flags (Y/N).<br>Description: Identify if the skill belongs to: DigComp – European Digital Competence Framework                                                                                                                                                                                         | NA | NA                        |
| skill_green      | Classification: Boolean flags (Y/N).<br>Description: Identify if the skill belongs to: Green Skills – environmentally relevant skills                                                                                                                                                                                          | NA | NA                        |
| skill_ict        | Classification: Boolean flags (Y/N).<br>Description: Identify if the skill belongs to: ICT Skills – information and communication technologies domain                                                                                                                                                                          | NA | NA                        |
| skill_skillstype | Classification: ESCO skill/competence type.<br>Description: Defines whether the ESCO entry is a skill/competence concept or a knowledge concept (skill type).<br>Values:<br>skill/competence<br>knowledge<br>Note: ESCO does not distinguish between "skill" and "competence" in practice; both are treated as skill concepts. | NA | NA                        |

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |    |                                 |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---------------------------------|
| skill_reusetype   | <p>Classification: ESCO reuse type.</p> <p>Description: Indicates whether the skill concept is originally defined by ESCO or reused from another classification/standard (e.g., ISCED-F, O*NET, DigComp).</p> <p>Values:</p> <p>ESCO-original</p> <p>reused (from external frameworks like ISCED, O*NET, DigComp, Canadian skill glossary, etc.)</p> <p>Purpose: Helps trace the origin and authority of the concept.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NA | NA                              |
| skill_description | <p>Classification: A description in ESCO is a text field providing a short explanation of the meaning of the concept and how it should be understood. Most importantly, it clarifies its semantic boundaries. For this reason, a description is always provided for each ESCO concept. It is sometimes followed with a scope note and/or a formal definition.</p> <p>Description: Short textual explanation of the skill/competence concept, defining its scope and intended meaning.</p> <p>Format: String (usually in multiple languages).</p> <p>Example: “The ability to prescribe medication to treat specific diseases.”</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | NA | NA                              |
| skill_language    | <p>Classification: ISO 639 language code (ESCO multilingual content).</p> <p>Description: Language of the skill label and description provided in ESCO.</p> <p>Format: Two-letter code (e.g. en, fr, de, ...).</p> <p>Note: ESCO provides translations in all 27 EU official languages.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | NA | NA                              |
| city_id           | <p>LAU (Local Administrative Units) code of the city indicated in the job advertisement.</p> <p>The LAUs are the building blocks of the NUTS (Nomenclature of territorial units for statistics <a href="https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Nomenclature_of_territorial_units_for_statistics_(NUTS))">https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Nomenclature_of_territorial_units_for_statistics_(NUTS))</a>, and comprise the municipalities and communes of the European Union:</p> <p>administrative for reasons such as the availability of data and policy implementation capacity; a subdivision of the NUTS 3 regions covering the whole economic territory of the Member States; appropriate for the implementation of local level typologies included in TERCET, namely the coastal area and DEGURBA classification.</p> <p><a href="https://ec.europa.eu/eurostat/web/nuts/local-administrative-units">https://ec.europa.eu/eurostat/web/nuts/local-administrative-units</a> The corresponding tables are published every year by Eurostat, since there are frequent changes to the LAUs. They can be downloaded here: <a href="https://ec.europa.eu/eurostat/web/nuts/local-administrative-units">https://ec.europa.eu/eurostat/web/nuts/local-administrative-units</a></p> |    | 2 to 9 digits<br>(alphanumeric) |
| city              | Name of the city (LAU) indicated in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    | string                          |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |                            |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|
| nuts3_id    | NUTS III region code version (2013, 2016 ??) indicated in the job advertisement.<br>NUTS III - small regions for specific diagnoses<br>European nomenclature of territorial units for statistics (NUTS):<br><a href="https://ec.europa.eu/eurostat/web/nuts/background">https://ec.europa.eu/eurostat/web/nuts/background</a><br>Definition:<br>In statistical context of Nomenclature of territorial units for statistics (NUTS), region refer to a geographical area: at sub-national level, a subdivision of a country, at different possible levels, but most commonly at level 1 of the NUTS. | NUTS III                | 5 digits<br>(alphanumeric) |
| nuts3       | Name of the NUTS III region inferred from the information in the job advertisement.<br>NUTS III - small regions for specific diagnoses                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NUTS III                | string                     |
| nuts2_id    | NUTS II region code indicated in the job advertisement.<br>NUTS II - basic regions for the application of regional policies                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NUTS II                 | 4 digits<br>(alphanumeric) |
| nuts2       | Name of the NUTS II region inferred from the information in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NUTS II                 | string                     |
| nuts1_id    | NUTS I region code indicated in the job advertisement.<br>NUTS I - major socio-economic regions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | NUTS I                  | 3 digits<br>(alphabetic)   |
| nuts1       | Name of the NUTS I region inferred from the information in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NUTS I                  | string                     |
| country_id  | Alpha-2 ISO code of the country where the job is to be performed ( <a href="https://www.iban.com/country-codes">https://www.iban.com/country-codes</a> ).<br>Computed based on the variable city.                                                                                                                                                                                                                                                                                                                                                                                                  | ISO<br>3166-1<br>alpha- | Alpha-2 code               |
| country     | Name of the country where the job is to be performed.<br>Computed based on the variable city.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | ISO<br>3166-1           | string                     |
| contract_id | Code of the type of contract offered in the job advertisement.<br>NLTD - Unlimited duration<br>LTD - Limited duration<br>SELF_EMP - Self-employed<br>APP_TRN - Apprentice or trainee                                                                                                                                                                                                                                                                                                                                                                                                               |                         | string                     |
| contract    | Type of contract offered in the job advertisement.<br>Definition:<br>An employment contract is a kind of contract used in labour law to attribute rights and responsibilities between an employee and an employer.                                                                                                                                                                                                                                                                                                                                                                                 |                         | string                     |

|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |                                                         |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------|
| education_id           | ISCED (International Standard Classification of Education) code of educational level required in the job advertisement.<br>ISCED 2011: <a href="https://ec.europa.eu/eurostat/statistics-explained/index.php?title=International_Standard_Classification_of_Education_(ISCED)#Implementation">https://ec.europa.eu/eurostat/statistics-explained/index.php?title=International_Standard_Classification_of_Education_(ISCED)#Implementation</a><br>Format : 1 digit (1 to 8, 0 - is not included)<br>Codes:<br>ED1 - Primary education<br>ED2 - Lower secondary education<br>ED3 - Upper secondary education<br>ED4 - Post-secondary non-tertiary education<br>ED5 - Short-cycle tertiary education<br>ED6 - Bachelor's or equivalent level<br>ED7 - Master's or equivalent level<br>ED8 - Doctoral or equivalent level | ISCED 2011  1 digit (1 to 8, 0 - is not included) |                                                         |
| education              | Educational level required in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | ISCED 2011                                        | string                                                  |
| economic_activity2d_id | NACE Rev2. section and division code.<br>Format: 1-digit alphabetical code or a two-digit numerical code.<br>Definition:<br>NACE Rev2. classification (from French: Nomenclature statistique des activités économiques dans la Communauté européenne):<br>an alphabetical code (sections), a two-digit numerical code (divisions), three-digit numerical code (groups), four-digit numerical code (classes).                                                                                                                                                                                                                                                                                                                                                                                                           | NACE Rev                                          | 1-digit alphabetical code or a two-digit numerical code |
| economic_activity2d    | Economic activity of the job advertisement. (NACE Rev2. section code)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | NACE Rev2                                         | string                                                  |
| economic_activity1d_id | NACE Rev2. division code.<br>Format: two-digit numerical code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | NACE Rev2                                         | two-digit numerical code                                |
| economic_activity1d    | Economic activity of the job advertisement. (NACE Rev2. division code).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | NACE Rev2                                         | string                                                  |
| activity_sector_id     | Code of the sector of economic activity (primary, secondary, tertiary), according to the industry classification.<br>A - primary: Agriculture, Forestry and Fishing<br>C - secondary: Manufacturing<br>G-U - tertiary: Services                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                   | string                                                  |
| activity_sector        | Sector of activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                   | string                                                  |

|                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| salary_id       | Code of the gross salary range offered, per year, in the job advertisement.<br>KE0-6 - From 0 to 6001 euros<br>KE6-12 - From 6001 to 12.000 euros<br>KE12-18 - From 12.001 to 18.000 euros<br>KE18-24 - From 18.001 to 24.000 euros<br>KE24-30 - From 24.001 to 30.000 euros<br>KE30-36 - From 30.001 to 36.000 euros<br>KE36-42 - From 36.001 to 42.000 euros<br>KE42-48 - From 42.001 to 48.000 euros<br>KE48-54 - From 48.001 to 54.000 euros<br>KE54-66 - From 54.001 to 66.000 euros<br>KE66-78 - From 66.001 to 78.000 euros<br>KE78-90 - From 78.001 to 90.000 euros<br>KE_GE90 - 90.001 euros or over | string |
| salary          | The label of the gross salary range offered in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | string |
| working_time_id | Code of the type of employment distinction - working time - sought/offered in the job advertisement.<br>PT - Part time<br>FT - Full time                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | string |
| working_time    | The label of type of employment distinction sought/offered in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | string |
| experience_id   | Code of years of work experience required in the job advertisement.<br>Y0_NE - No experience<br>Y_LT1 - Less than 1 year<br>Y1-2 - From 1 to 2 years<br>Y2-4 - From 2 to 4 years<br>Y4-6 - From 4 to 6 years<br>Y6-8 - From 6 to 8 years<br>Y8-10 - From 8 to 10 years<br>Y_GT10 - Over 10 years                                                                                                                                                                                                                                                                                                              | string |
| experience      | The number of years of work experience required in the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | string |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |              |
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| source_category_id | Code of the category of the source where the job advertisement was published - with the following codes:<br>EMP_ORG - Employers' organization<br>COMP_WEB - Company website<br>EDSITE - Education sites<br>EMPAG_WEB - Employment website (employment agency working only online)<br>JOBSE - Job search engine (websites collecting OJA from different sources and presenting them in an integrated search engine)<br>SNETW - Social Network<br>PEA - Private Employment Agency (physical Employment agency website (i.e. Adecco); usually works on both direct and web channels)<br>ONEWSP - Online newspaper (advertisement pages of online newspapers)<br>PES - Public employment service (portals of national employment services)<br>CL_ADSP - Classified ads portal (general ads portals containing ads for job vacancies) |                    | string       |
| source_category    | Name of the category of the source where the job advertisement was published.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                    | string       |
| source_country     | The Alpha-2 ISO code of the country where the source website providing the job advertisement is hosted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | ISO 3166-1 alpha-2 | Alpha-2 code |
| source_id          | Code id (anonymized) of the job portal (or website) where the job advertisement is retrieved from.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                    | string       |
| source             | The job portal (or website) name where the job advertisement is retrieved from. In ingestion monitoring, the source is the provider of the job advertisement collected through direct access, scraping or crawling; it is the website where the job advertisement is indexed. A source could redirect to another website where the job advertisement is actually published: therefore, the site is the website where the job advertisement is downloaded. The source and the site of a job ad does not necessarily coincide.                                                                                                                                                                                                                                                                                                     |                    | string       |
| site               | The website name where the job is published, i.e. the aggregator portal is the source of the job advertisement, the company website is the site where the job is published.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                    | string       |
| company            | The self-reported name of the entity - usually company or staffing agency - that published the job advertisement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                    | string       |

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                        |
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| source_stability | <p>Indicates if the source from which the advertisement was extracted is "stable", "unstable" or a "new source" (not yet classified). Stable source: definition and methodology The goal of the source stability analysis is to identify sources that provide data in a stable and reliable way over time; these sources can be used to calculate time series of indicators based on OJA data that are comparable over time. Issues to be addressed: High number of sources to be managed: over 400 different sources. The stability of a source can be different in different countries. Some of the sources are considered not stable over time due to: - Technical problems related to data ingestion - Source preventing the data-scraping algorithms from collecting data - Technical problems of the source (web site non available) - Lifecycle of the source (new sources, closed sources, acquisitions) Source stability identification: We developed an algorithm to identify/flag unstable source-country pairs, so that the analyst can decide whether to use them or not. We recommend using all sources in the dataset for descriptive, cross-sectional analysis, and to use only stable sources when dealing with time series comparisons. Moreover, the source evaluation may vary over time: the algorithm is based on a time window so that the list of stable sources can change over time adapting to scope of the analysis. Steps of the algorithm: - Analysis at source country level in a specific time-frame. - Problem detection: Coverage, Variability, Outliers, Irregular trends. - Categorization: Stable, Unstable, Newcomer. Source-country pairs are considered unstable if they do not meet the following criteria: - Coverage (number of months with observations over the number of months in the time window): <math>&lt; 80\%</math> - Index Coverage (number of months with observations over the number of months since the source was introduced): <math>&lt; 80\%</math> - Variability (coefficient of variation): <math>&gt; 0.6</math> - Outliers (<math>IQR = Q3 - Q1</math>): <math>&lt; \text{lower range } (Q1 - (3 * IQR))</math> or <math>&gt; \text{upper range } (Q1 + (3 * IQR))</math> - Irregular trends (average number of job postings of the first quarter of the last years is compared; if the difference (+/-) is too large, the pair source-country is flagged as irregular): <math>&lt; -100\%</math> or <math>&gt; 100\%</math> In the future, as the number of available months increases, it will be possible to implement a sensitivity analysis of the source stability algorithm. Following the alerts, every source-country pair is categorized as follows: - stable (source-country with no alerts). - newcomer (new source-country for which the stability indicators could not be calculated</p> | . - unstable (all t e rest).    string |
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## 6 Version info / generation timestamp

Report generated on: 2026-07-23